

21.6 Autoregressive errors

Serial correlation raises the same choice as heteroscedasticity: model it and use generalized least squares, or keep least squares and estimate its covariance without a model. Either way dependence must be described by few parameters, and the simplest description, the one behind the Durbin–Watson test, is the first-order autoregression.

21.6.1 The AR(1) model

Definition 21.6.1. First-order autoregressive errors

The errors follow a **first-order autoregressive process**, AR(1), if

$$\varepsilon_t = \rho \varepsilon_{t-1} + u_t, \quad t = 2, \dots, n,$$

with $|\rho| < 1$, where $u_2, \dots, u_n$ are uncorrelated with mean zero and variance σ_u^2 , and the starting value ε_1 has mean zero, variance $\sigma_u^2/(1 - \rho^2)$ and is uncorrelated with the u_t .

The variance of ε_1 is the one that makes the process **stationary**: it is the variance an AR(1) process reaches after running for a long time, and with it every ε_t has the same variance.

Theorem 21.6.1. Regression with AR(1) errors

Let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\text{rank}(\mathbf{X}) = p$ and AR(1) errors.

- (a) $\text{Cov}(\varepsilon_s, \varepsilon_t) = \sigma_u^2 \rho^{|s-t|}/(1 - \rho^2)$. Write $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma_u^2 \mathbf{V}_\rho$.
- (b) (*Prais–Winsten transformation.*) Let $\mathbf{L}_\rho$ be the $n \times n$ lower bidiagonal matrix with $(\mathbf{L}_\rho)_{11} = \sqrt{1 - \rho^2}$, $(\mathbf{L}_\rho)_{tt} = 1$ and $(\mathbf{L}_\rho)_{t,t-1} = -\rho$ for $t \geq 2$. Then $\text{Cov}(\mathbf{L}_\rho \boldsymbol{\varepsilon}) = \sigma_u^2 \mathbf{I}$, so $\mathbf{V}_\rho^{-1} = \mathbf{L}_\rho^\top \mathbf{L}_\rho$ and $\det \mathbf{V}_\rho = 1/(1 - \rho^2)$. For known ρ , the generalized least squares estimate is ordinary least squares on $(\mathbf{L}_\rho \mathbf{y}, \mathbf{L}_\rho \mathbf{X})$, and it is the best linear unbiased estimator.
- (c) (*Likelihood.*) If the errors are normal, the log-likelihood is

$$\begin{aligned} \ell(\boldsymbol{\beta}, \rho, \sigma_u^2) = & -\frac{n}{2} \log(2\pi\sigma_u^2) + \frac{1}{2} \log(1 - \rho^2) \\ & - \frac{\|\mathbf{L}_\rho(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\|^2}{2\sigma_u^2}. \end{aligned}$$

- (d) (*Cochrane–Orcutt.*) Let $S(\boldsymbol{\beta}, \rho) = \sum_{t=2}^n \{(y_t - \rho y_{t-1}) - (\mathbf{x}_{(t)} - \rho \mathbf{x}_{(t-1)})^\top \boldsymbol{\beta}\}^2$, the sum of squares of the last $n - 1$ rows of $\mathbf{L}_\rho(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$. Alternate the steps

$$\begin{aligned} \rho_{k+1} &= \frac{\sum_{t \geq 2} e_t e_{t-1}}{\sum_{t \geq 2} e_{t-1}^2} \quad \text{with } \mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}_k, \\ \boldsymbol{\beta}_{k+1} &= \arg \min_{\mathbf{b}} S(\mathbf{b}, \rho_{k+1}). \end{aligned}$$

Then $S(\boldsymbol{\beta}_{k+1}, \rho_{k+1}) \leq S(\boldsymbol{\beta}_k, \rho_{k+1}) \leq S(\boldsymbol{\beta}_k, \rho_k)$, so S decreases along the iterations, and any limit point $(\boldsymbol{\beta}^*, \rho^*)$ with $|\rho^*| < 1$ at which the steps are continuous is a stationary point of S .

Proof.

- (a) By induction $\text{Var } \varepsilon_t = \sigma_u^2/(1 - \rho^2)$ for all t , because $\text{Var } \varepsilon_t = \rho^2 \text{Var } \varepsilon_{t-1} + \sigma_u^2$, ε_{t-1} being a linear combination of $\varepsilon_1, u_2, \dots, u_{t-1}$ and so uncorrelated with u_t . For $k \geq 1$, iterating the recursion gives $\varepsilon_t = \rho^k \varepsilon_{t-k} + \sum_{j=0}^{k-1} \rho^j u_{t-j}$, and the sum is uncorrelated with ε_{t-k} , so $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = \rho^k \text{Var } \varepsilon_{t-k}$.
- (b) $\mathbf{L}_\rho \boldsymbol{\varepsilon} = (\sqrt{1 - \rho^2} \varepsilon_1, u_2, \dots, u_n)^\top$, whose entries are uncorrelated with variance σ_u^2 . So $\sigma_u^2 \mathbf{L}_\rho \mathbf{V}_\rho \mathbf{L}_\rho^\top = \sigma_u^2 \mathbf{I}$, and since $\mathbf{L}_\rho$ is nonsingular, $\mathbf{V}_\rho^{-1} = \mathbf{L}_\rho^\top \mathbf{L}_\rho$. $\mathbf{L}_\rho$ is triangular with determinant $\sqrt{1 - \rho^2}$, so $\det \mathbf{V}_\rho = 1/(1 - \rho^2)$. The last statement is Theorem 6.9.1 and Corollary 7.2.3 with the whitening matrix $\mathbf{L}_\rho$, as in Section 6.9.
- (c) The normal density of $\boldsymbol{\varepsilon}$ with covariance $\sigma_u^2 \mathbf{V}_\rho$ has $-\frac{1}{2} \log \det(\sigma_u^2 \mathbf{V}_\rho) = -\frac{n}{2} \log \sigma_u^2 + \frac{1}{2} \log(1 - \rho^2)$ and quadratic form $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}_\rho^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) / \sigma_u^2 = \|\mathbf{L}_\rho(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\|^2 / \sigma_u^2$.
- (d) For fixed $\boldsymbol{\beta}$, $S(\boldsymbol{\beta}, \rho) = \sum_{t \geq 2} (e_t - \rho e_{t-1})^2$ is a quadratic in ρ minimized by ρ_{k+1} , which gives the second inequality. For fixed ρ , S is a least squares criterion in $\mathbf{b}$, minimized by $\boldsymbol{\beta}_{k+1}$, which gives the first. At a limit point both steps leave $(\boldsymbol{\beta}^*, \rho^*)$ unchanged, so $\partial S / \partial \rho = 0$ and $\partial S / \partial \boldsymbol{\beta} = \mathbf{0}$ there. ■

The whitening in (b) is the transformation of Prais and Winsten (1954). Row $t \geq 2$ of the transformed model is the **quasi-difference** $y_t - \rho y_{t-1} = (\mathbf{x}_{(t)} - \rho \mathbf{x}_{(t-1)})^\top \boldsymbol{\beta} + u_t$, and row one is the first observation rescaled by $\sqrt{1 - \rho^2}$. The inverse covariance $\mathbf{V}_\rho^{-1}$ is tridiagonal, with diagonal $(1, 1 + \rho^2, \dots, 1 + \rho^2, 1)$ and off-diagonal entries $-\rho$ (Exercise (A1)), which is why AR(1) models are cheap to fit however long the series. Cochrane and Orcutt (1949) dropped the first row, which leaves the conditional sum of squares S of (d). Dropping it loses little when the regressors are stationary, but it can lose a great deal when they trend, because the first observation is then the only one whose transformed regressor is not nearly constant. Maximizing the likelihood of (c) keeps the first row and the term $\frac{1}{2} \log(1 - \rho^2)$, which keeps $\hat{\rho}$ inside $(-1, 1)$.

21.6.2 Feasible generalized least squares

In practice ρ is unknown. The two-step **feasible** estimator estimates it from the ordinary least squares residuals, $\hat{\rho} = \sum_{t \geq 2} \hat{\varepsilon}_t \hat{\varepsilon}_{t-1} / \sum_{t \geq 2} \hat{\varepsilon}_{t-1}^2$, and then applies the Prais–Winsten transformation with $\hat{\rho}$. Iterating the two steps gives either the Cochrane–Orcutt estimate or, keeping the first row, a close relative of the maximum likelihood estimate.

Remark (What is known about feasible GLS with AR(1) errors). Suppose the errors are a stationary AR(1) process with independent innovations of finite variance, and the regressors are stationary, or have polynomial trends, with suitable limits for their sample moments. Then $\hat{\rho} \rightarrow \rho$ in probability, and $\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{FGLS}} - \hat{\boldsymbol{\beta}}_{\text{GLS}}) \rightarrow \mathbf{0}$ after the appropriate normalization for trending regressors, so the feasible estimator has the limiting distribution of the optimal one and the usual standard errors from the transformed regression are valid in large samples. We do not prove this. A sketch: $\hat{\rho}$ differs from the sample autocorrelation of the true errors by terms bounded by multiples of $\|\mathbf{M}\boldsymbol{\varepsilon}\| \|\boldsymbol{\varepsilon}\|/n$ and $\|\mathbf{M}\boldsymbol{\varepsilon}\|^2/n$, which tend to zero because $\|\mathbf{M}\boldsymbol{\varepsilon}\|$ is bounded in probability (compare the proof of Theorem 21.2.2(c)), and that autocorrelation is consistent by a law of large numbers for stationary sequences; the comparison of $\hat{\boldsymbol{\beta}}_{\text{FGLS}}$ with $\hat{\boldsymbol{\beta}}_{\text{GLS}}$ then follows the pattern of Theorem 21.3.1(d), because $\mathbf{L}_\rho$ depends smoothly on

ρ . Fuller (1996) gives complete statements and proofs. The finite-sample behaviour must be checked by simulation.

Example. Okun's law with AR(1) errors

For the regression of Example (A quarterly Okun's-law regression), $\hat{\rho} = 0.303$. The two-step Prais–Winsten estimate of the slope is -0.0506 with standard error 0.0049 , against the least squares -0.0671 with nominal standard error 0.0050 . Iterating Cochrane–Orcutt to convergence takes 17 iterations and ends at $\hat{\rho} = 0.548$ and slope -0.0380 . The estimate of the effect of growth on unemployment thus changes by almost half depending on the treatment of the errors, and the residuals of the transformed regression still have lag-one autocorrelation 0.148 . Both facts suggest that the AR(1) error model does not describe these data. We return to this below.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

macro = sm.datasets.macrodatab.load_pandas().data
du = np.diff(macro["unemp"].to_numpy())
growth = 400 * np.diff(np.log(macro["realgdp"].to_numpy()))
n = len(du)
X = np.column_stack([np.ones(n), growth])

def prais_winsten(X, y, rho):
    """Whiten AR(1) errors: row 1 times sqrt(1 - rho^2), row t - rho row t-1."""
    Xs, ys = X.copy(), y.copy()
    Xs[1:], ys[1:] = X[1:] - rho * X[:-1], y[1:] - rho * y[:-1]
    Xs[0], ys[0] = np.sqrt(1 - rho**2) * X[0], np.sqrt(1 - rho**2) * y[0]
    return Xs, ys

def ols(X, y):
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
    e = y - X @ b
    cov = (e @ e / (len(y) - X.shape[1])) * np.linalg.inv(X.T @ X)
    return b, e, cov

b_ols, e, V_ols = ols(X, du)
# regress e_t on e_(t-1)
rho = np.sum(e[1:] * e[:-1]) / np.sum(e[:-1]**2)
# two-step Prais-Winsten
b_pw, e_pw, V_pw = ols(*prais_winsten(X, du, rho))
print(f"rho-hat {rho:.3f}")
print(f"OLS slope {b_ols[1]:.4f} (se {np.sqrt(V_ols[1, 1]):.4f}); "
      f"Prais-Winsten {b_pw[1]:.4f} (se {np.sqrt(V_pw[1, 1]):.4f})")
```

[Run this code online](#)

```
def cond_ss(b, r):
    """Conditional sum of squares S(beta, rho) over t = 2, ..., n."""
    u = (du[1:] - r * du[:-1]) - (X[1:] - r * X[:-1]) @ b
```

```

return u @ u

b_co, r_co, path = b_ols, 0.0, []
for it in range(100):
    res = du - X @ b_co
    r_new = np.sum(res[1:] * res[:-1]) / np.sum(res[:-1] ** 2)    # rho step
    path.append(cond_ss(b_co, r_new))
    # beta step
    b_co, *_ = np.linalg.lstsq(X[1:] - r_new * X[:-1], du[1:] - r_new * du[:-1],
                               rcond=None)
    path.append(cond_ss(b_co, r_new))
    if abs(r_new - r_co) < 1e-10:
        break
    r_co = r_new
print(f"Cochrane-Orcutt after {it + 1} iterations: rho {r_co:.4f}, "
      f"slope {b_co[1]:.4f}")

```

[Run this code online](#)

21.6.3 Efficiency and validity of least squares

How much is lost by ignoring AR(1) errors? There are two separate questions: how much less efficient least squares is than generalized least squares, and how wrong its nominal standard error $s\{(\mathbf{X}^\top \mathbf{X})^{-1}\}_{jj}^{1/2}$ is. The answers depend on the regressor, and they run in opposite directions.

Example. Three regressors

Take $n = 100$, an intercept, and one of three regressors: a linear trend, a fixed draw of white noise, and a fixed draw of an AR(1) series with coefficient 0.8 (its sample lag-one autocorrelation is 0.87). For AR(1) errors, [Figure 21.6.1](#) shows (a) the ratio of the generalized to the ordinary least squares variance of the slope, and (b) the ratio of the true least squares variance to the expected nominal one, $E(s^2)\{(\mathbf{X}^\top \mathbf{X})^{-1}\}_{22}$, with $E s^2$ from Theorem 2.3.1. All are exact computations.

At $\rho = 0.6$: for the trend, least squares is 0.970 efficient but its nominal variance is too small by a factor of 4.01. For white noise it is only 0.462 efficient, yet its nominal variance is nearly right (factor 1.11). The autocorrelated regressor suffers both ways: efficiency 0.650, variance understated by a factor of 3.19. At $\rho = 0.9$ the factors for the trend become 0.842 and 19.59.

The trend repeats the lesson of the autocorrelated-errors example of [Section 19.2](#); the other two regressors are new.

The pattern has a simple explanation. The variance of $\hat{\beta}_1$ in a simple regression is $\sum_{s,t} (x_s - \bar{x})(x_t - \bar{x}) \text{Cov}(\varepsilon_s, \varepsilon_t) / S_{xx}^2$. When the regressor is itself autocorrelated, neighbouring terms add up, and for a stationary regressor with lag-one autocorrelation ϕ the ratio of true to nominal variance is approximately $(1 + \rho\phi)/(1 - \rho\phi)$ ([Exercise \(B2\)](#)). For the AR(0.8) regressor this gives 3.18 at $\rho = 0.6$, against the exact 3.19. A white-noise regressor has $\phi \approx 0$, so the nominal variance is nearly right; but generalized least squares, which quasi-differences x_t , turns a rough regressor into an even rougher one with more information, and least squares' efficiency tends to $(1 - \rho^2)/(1 + \rho^2)$, which is 0.471 at $\rho = 0.6$ ([Exercise \(B1\)](#)). For polynomial trends, Grenander (1954) showed that least squares is asymptotically fully efficient, which the trend curve in panel (a) approaches; but its standard error is then at its most misleading.

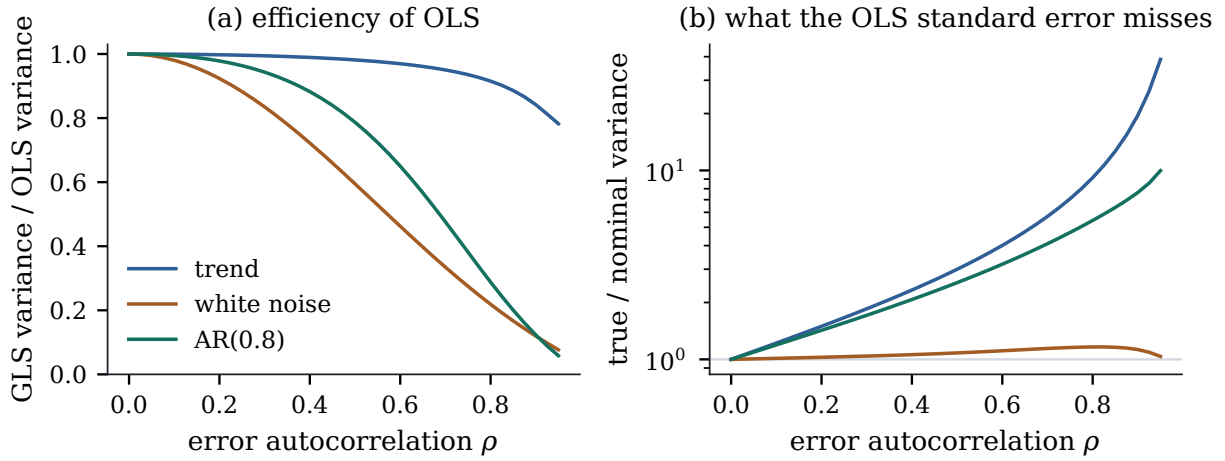


Figure 21.6.1: Least squares with AR(1) errors, $n = 100$, for three regressors. (a) Efficiency: variance of the generalized least squares slope divided by that of the ordinary one. (b) True variance of the least squares slope divided by its expected nominal variance (log scale).

21.6.4 Newey–West standard errors

The analogue of the sandwich of Section 21.4 for serially correlated errors keeps least squares and estimates the middle matrix $\mathbf{X}^\top \text{Cov}(\boldsymbol{\varepsilon})\mathbf{X} = \sum_{s,t} \text{Cov}(\varepsilon_s, \varepsilon_t) \mathbf{x}_{(s)} \mathbf{x}_{(t)}^\top$ without a parametric model. The products $\hat{\varepsilon}_s \hat{\varepsilon}_t$ can stand in for the covariances only when s and t are close, since distant covariances are small but their estimates are not. With $\mathbf{v}_t = \mathbf{x}_{(t)} \hat{\varepsilon}_t$ and a truncation lag L , Newey and West (1987) proposed

$$\begin{aligned} \hat{\boldsymbol{\Omega}}_L &= \sum_{k=-L}^L \left(1 - \frac{|k|}{L+1}\right) \hat{\mathbf{\Gamma}}_k, \\ \hat{\mathbf{\Gamma}}_k &= \sum_{t=k+1}^n \mathbf{v}_t \mathbf{v}_{t-k}^\top, \quad \hat{\mathbf{\Gamma}}_{-k} = \hat{\mathbf{\Gamma}}_k^\top, \end{aligned} \tag{21.6.1}$$

and the covariance estimate $(\mathbf{X}^\top \mathbf{X})^{-1} \hat{\boldsymbol{\Omega}}_L (\mathbf{X}^\top \mathbf{X})^{-1}$. The declining **Bartlett weights** $1 - |k|/(L+1)$ are not a matter of taste.

Proposition 21.6.2. Newey–West estimates are nonnegative definite

For any vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ and any $L \geq 0$, $\hat{\boldsymbol{\Omega}}_L$ in Eq. 21.6.1 is nonnegative definite. Without the weights, $\sum_{|k| \leq L} \hat{\mathbf{\Gamma}}_k$ need not be.

Proof. Set $\mathbf{v}_t = \mathbf{0}$ for $t \leq 0$ and $t > n$, and let $\mathbf{s}_j = \sum_{i=0}^L \mathbf{v}_{j-i}$ for every integer j . Then

$$\sum_j \mathbf{s}_j \mathbf{s}_j^\top = \sum_j \sum_{i,l=0}^L \mathbf{v}_{j-i} \mathbf{v}_{j-l}^\top = \sum_{k=-L}^L (L+1 - |k|) \sum_t \mathbf{v}_t \mathbf{v}_{t-k}^\top,$$

because for fixed $t = j - i$ and lag $k = l - i$ there are exactly $L+1 - |k|$ pairs (i, l) in $\{0, \dots, L\}^2$ with $l - i = k$. Dividing by $L+1$ shows that $\hat{\boldsymbol{\Omega}}_L = (L+1)^{-1} \sum_j \mathbf{s}_j \mathbf{s}_j^\top$, a sum of

nonnegative definite matrices. [Exercise \(C1\)](#) gives an example for the unweighted sum. ■

Newey and West (1987) proved that the estimator is consistent when L grows with n slowly enough, for errors and regressors that are stationary with dependence dying out fast enough. We do not prove this. Its practical weakness is the choice of L , which trades bias from truncation against variance from estimating many autocovariances; Andrews (1991) studied data-based choices. A common rule of thumb is $L = \lfloor 4(n/100)^{2/9} \rfloor$. For the Okun regression, with $L = 4$, the Newey–West standard error of the slope is 0.0086, against the nominal 0.0050. With $L = 0$ the estimator is HC0 ([Exercise \(A2\)](#)).

```
def newey_west(X, e, L):
    """Bartlett-weighted HAC covariance of the OLS estimate, truncation lag L."""
    v = X * e[:, None]
    S = v.T @ v
    for k in range(1, L + 1):
        G = v[k:].T @ v[:-k]
        S += (1 - k / (L + 1)) * (G + G.T)
    B = np.linalg.inv(X.T @ X)
    return B @ S @ B

V_nw = newey_west(X, e, 4)
print(f"Newey-West (L = 4) se of slope {np.sqrt(V_nw[1, 1]):.4f}")
```

[Run this code online](#)

Example. Coverage with AR(1) errors

Let the regressor be a fixed draw of an AR(1) series with coefficient 0.8, and the errors AR(1) with $\rho = 0.6$ and normal innovations. Over 10,000 simulated data sets the nominal 95% intervals for the slope covered as follows. For $n = 50$: least squares with the usual standard error 0.747, least squares with Newey–West 0.784, two-step Prais–Winsten 0.913, and generalized least squares with the true ρ 0.946. For $n = 200$: 0.746, 0.871, 0.944 and 0.950. The Newey–West interval is valid only asymptotically, and converges slowly; when the AR(1) model is right, using it pays.

```
def ar1_draws(n, rho, reps, rng):
    u = rng.normal(size=(reps, n))
    x = np.empty((reps, n))
    x[:, 0] = u[:, 0] / np.sqrt(1 - rho**2)
    for t in range(1, n):
        x[:, t] = rho * x[:, t - 1] + u[:, t]
    return x

def coverage(n, reps, rng):
    """Coverage of 95% slope intervals: OLS, Newey-West, Prais-Winsten, GLS."""
    x = ar1_draws(n, 0.8, 1, rng)[0]
    Xc = np.column_stack([np.ones(n), x])
    L = int(4 * (n / 100) ** (2 / 9))
    q = stats.t.ppf(0.975, n - 2)
    hits = np.zeros(4)
```

```

for y in ar1_draws(n, 0.6, reps, rng):          # true slope 0
    b, res, V = ols(Xc, y)
    r = np.sum(res[1:] * res[:-1]) / np.sum(res[:-1] ** 2)
    b_f, _, V_f = ols(*prais_winsten(Xc, y, r))
    b_g, _, V_g = ols(*prais_winsten(Xc, y, 0.6))
    ses = [V[1, 1], newey_west(Xc, res, L)[1, 1], V_f[1, 1], V_g[1, 1]]
    hits += [abs(bb[1]) <= q * np.sqrt(v)
              for bb, v in zip((b, b_f, b_g), ses)]
return hits / reps

rng = np.random.default_rng(2109)
for n_sim in (50, 200):
    print(n_sim, "OLS, Newey-West, Prais-Winsten, GLS:",
          np.round(coverage(n_sim, 500, rng), 3))

```

[Run this code online](#)

21.6.5 Serial correlation as a symptom

An AR(1) error model is a restriction on a dynamic regression. Multiply the model at time $t - 1$ by ρ and subtract:

$$y_t = \rho y_{t-1} + \mathbf{x}_{(t)}^\top \boldsymbol{\beta} - \rho \mathbf{x}_{(t-1)}^\top \boldsymbol{\beta} + u_t.$$

This is a regression of y_t on its own lag and on current and lagged regressors, with the coefficients of the lagged regressors constrained to equal $-\rho$ times those of the current ones, the **common factor restriction**. Residual autocorrelation is often the trace of a missing lag of y or of a regressor, and then the dynamic regression without the restriction is the better model. Hendry and Mizon (1978) argued that the restriction should be tested before the error model is adopted.

Example. Okun's law, dynamically

Regressing Δu_t on an intercept, Δu_{t-1} , g_t and g_{t-1} gives coefficients 0.2160, 0.3571 (standard error 0.0574), -0.0487 (0.0043) and -0.0162 (0.0055). The AR(1) model would require the last coefficient to be $-0.3571 \times (-0.0487) = 0.0174$. The estimate has the opposite sign, and the delta-method test of the restriction gives $z = -8.1$. The unrestricted dynamic regression has no remaining residual autocorrelation: its Breusch–Godfrey statistic with four lags is 4.42, with p-value 0.35. Unemployment responds to growth with a lag and with persistence; the AR(1) error model mistook this for correlated noise, and the slope estimates of [Example \(Okun's law with AR\(1\) errors\)](#), which move with $\hat{\rho}$, were artefacts of that mistake.

```

# the AR(1) error model is the dynamic regression
#   du_t = a + rho du_(t-1) + b g_t + c g_(t-1) + u_t
#   with the restriction c = -rho b
Xd = np.column_stack([np.ones(n - 1), du[:-1], growth[1:], growth[:-1]])
bd, ed, Vd = ols(Xd, du[1:])
# zero under the restriction
restr = bd[3] + bd[1] * bd[2]
grad = np.array([0.0, bd[2], bd[1], 1.0])          # delta method
z_cf = restr / np.sqrt(grad @ Vd @ grad)
print("dynamic regression: const, du(t-1), g(t), g(t-1) =", np.round(bd, 4))
print(f"common-factor restriction c + rho b = {restr:.4f}, z = {z_cf:.1f}")

```

[Run this code online](#)

An extreme case is the regression of one random walk on another, independent one, which gives large t statistics and a small d far more often than the nominal level suggests (Granger and Newbold 1974). There the residual autocorrelation signals a meaningless regression, and no correction of the standard errors repairs it.

21.6.6 Exercises

21.6.6.1 A. Check your understanding

Exercise. A1

Compute $\mathbf{L}_\rho^\top \mathbf{L}_\rho$ and show that it is tridiagonal with diagonal $(1, 1 + \rho^2, \dots, 1 + \rho^2, 1)$ and off-diagonal entries $-\rho$. Check $\mathbf{V}_\rho \mathbf{L}_\rho^\top \mathbf{L}_\rho = \mathbf{I}$ for $n = 2$.

Solution. Column t of $\mathbf{L}_\rho$ has entry 1 (or $\sqrt{1 - \rho^2}$ for $t = 1$) in row t and $-\rho$ in row $t + 1$ (for $t < n$). The inner product of columns t and t is $1 + \rho^2$ for $1 < t < n$, $(1 - \rho^2) + \rho^2 = 1$ for $t = 1$ and 1 for $t = n$; columns t and $t + 1$ share only row $t + 1$, giving $-\rho$; other pairs share nothing. For $n = 2$, $\mathbf{V}_\rho = (1 - \rho^2)^{-1} \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ and $\mathbf{L}_\rho^\top \mathbf{L}_\rho = \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$, whose product is $(1 - \rho^2)^{-1}(1 - \rho^2)\mathbf{I} = \mathbf{I}$.

Exercise. A2

Show that the Newey–West estimator with $L = 0$ is HC0, and that with $\mathbf{X} = \mathbf{1}$ and $L \geq 1$ it estimates the long-run variance $\sum_k \gamma_k$ of a stationary series, scaled by n .

21.6.6.2 B. Practice

Exercise. B1

Let the regressor x_t be white noise with variance τ^2 , independent of AR(1) errors, and consider the slope in a regression through the origin. Show heuristically that $n \text{Var}(\hat{\beta}_{\text{OLS}}) \rightarrow \sigma_u^2 / \{(1 - \rho^2)\tau^2\}$ and $n \text{Var}(\hat{\beta}_{\text{GLS}}) \rightarrow \sigma_u^2 / \{(1 + \rho^2)\tau^2\}$, so that the efficiency of least squares, the ratio bounded in Theorem 19.2.3, tends to $(1 - \rho^2)/(1 + \rho^2)$.

Exercise. B2

Let the regressor be a stationary AR(1) series with coefficient ϕ , independent of AR(1) errors with coefficient ρ . Show heuristically that the ratio of the true variance of the least squares slope to its nominal variance tends to $(1 + \rho\phi)/(1 - \rho\phi)$.

Solution. Conditionally on the regressor, $\text{Var}(\hat{\beta}) = \sum_{s,t} \tilde{x}_s \tilde{x}_t \gamma_0 \rho^{|s-t|} / S_{xx}^2$, where $\tilde{x}_t = x_t - \bar{x}$. For large n , $\sum_t \tilde{x}_t \tilde{x}_{t-k} \approx S_{xx} \phi^{|k|}$, so the variance is about $(\gamma_0 / S_{xx}) \sum_k (\rho\phi)^{|k|} = (\gamma_0 / S_{xx})(1 + \rho\phi)/(1 - \rho\phi)$. The nominal variance $E(s^2)/S_{xx}$ is about γ_0 / S_{xx} , since s^2 estimates the marginal error variance γ_0 when p/n is small.

Exercise. B3

In the model $y_t = \alpha + \beta t + \varepsilon_t$, $t = 1, \dots, n$, with AR(1) errors, show that the Cochrane–Orcutt regression has columns $(1 - \rho)\mathbf{1}$ and $(1 - \rho)t + \rho$ on $t = 2, \dots, n$, and that the information it carries about β is $(1 - \rho)^2 \sum_{t \geq 2} (t - \bar{t})^2 / \sigma_u^2$, where $\bar{t}$ is the mean of $2, \dots, n$. What happens as $\rho \rightarrow 1$ with n fixed, and why does the Prais–Winsten first row change the picture?

21.6.6.3 C. Going deeper**Exercise. C1**

Give vectors v_1, v_2, v_3 (with $p = 1$) for which the unweighted sum $\sum_{|k| \leq 1} \hat{\Gamma}_k$ is negative.

Solution. Take $(v_1, v_2, v_3) = (1, -1, 1)$. Then $\hat{\Gamma}_0 = 3$ and $\hat{\Gamma}_1 = \hat{\Gamma}_{-1} = v_2 v_1 + v_3 v_2 = -2$, so the unweighted sum is $3 - 4 = -1$. With Bartlett weights it is $3 - 2 = 1$.